

Database Constraints

① Domain constraint

<u>ID</u>	name
01	Arib

id int;

INSERT <'ab', 'Rahman'>

⊗

Primary { NULL ⊗ — Entity integrity constraint
UNIQUE ✓ — Key

② Key constraint → Primary Key unique হইবে

<u>ID</u>	name
1	a
2	b

<'2', 'Ahmed'> ⊗

<'3', 'b'> ✓

③ Entity integrity constraint

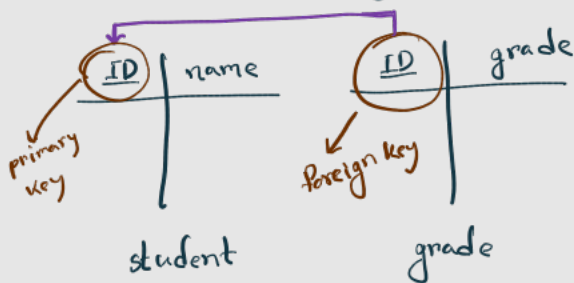
↳ Primary Key NULL

<u>ID</u>	name
NULL	

⊗

④ Referential Integrity constrain

F.K → Primary



(a)

<u>ID</u>	name	<u>ID</u>	Grade
1		1	
2		3	

* Foreign Key এর মান Primary Key এর মানের সমান হওয়া উচিত

⊗

(b)

<u>ID</u>	name	<u>ID</u>	Grade
1		1	
2		2	
3			

✓

(c)

<u>ID</u>	name	<u>ID</u>	Grade
1		1	
2		2	
3			

<u>ID</u>	name	<u>ID</u>	Grade
1		1	
2		2	
3			

EMPLOYEE									
Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Yess, Houston, TX	M	40000	333445555	5
Alicia	J	Zelaya	999985777	1968-01-19	3321 Castle, Spring, TX	F	43000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	333445555	4
Ramesh	K	Namesh	666664444	1962-09-15	975 Fire Cak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-21	5631 Riiss, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987997897	1969-03-20	990 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888666555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT			
Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS	
Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON		
Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453433453	1	20.0
453433453	2	20.0
333445555	3	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999987777	30	30.0
999987777	10	10.0
987987987	10	35.0
987997987	20	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT			
Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	3
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT				
Essn	Dependent_name	Sex	Bdate	Relationship
333445559	Alice	F	1986-04-06	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

① Insert <D1, standford>
in DEPT_Locations

Domain constraint
violate,

② Insert <6, standford>
in DEPT_Locations

Referential I.C
Reason: P.K 6 to 6 नहीं

③ Insert <null, standford>
in DEPT_Locations

Entity I.C

④ Delete Franklin row from employee table

Referential I.C

⑤ Update ProductX Pnumber to 20

Key constraint